

FRAUD REPORT

Dataset : SYNTHETIC FINANCIAL DATA

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SUMMARY

1. Business Problem
2. Dataset
3. Analysis
4. Key Findings
5. Recommendations

1. Business Problem

A financial institution has observed fraudulent transactions within its payment system. The objective is to identify which customer segments, transaction categories, locations, and times of day exhibit elevated fraud risk so the fraud operations team can prioritize monitoring and allocate investigative resources more effectively.

2. Dataset

- A. Number of transactions : 10000 transactions
- B. Number of features : There are 11 features
- C. Transaction period : 2023-01-01 from 00:00:00 to 02:00:00
- D. Key variables : amount, is_fraudulent, card_type, location, purchase category, customer_age

3. Methodology

Data preparation:

- a. Imported the CSV file into python using pandas
- b. Converted transaction data stamp into datetime format
- c. Extracted the transaction hour to enable time based-analysis
- d. Verified data quality by checking for missing values and duplicate records

Exploratory Data Analysis:

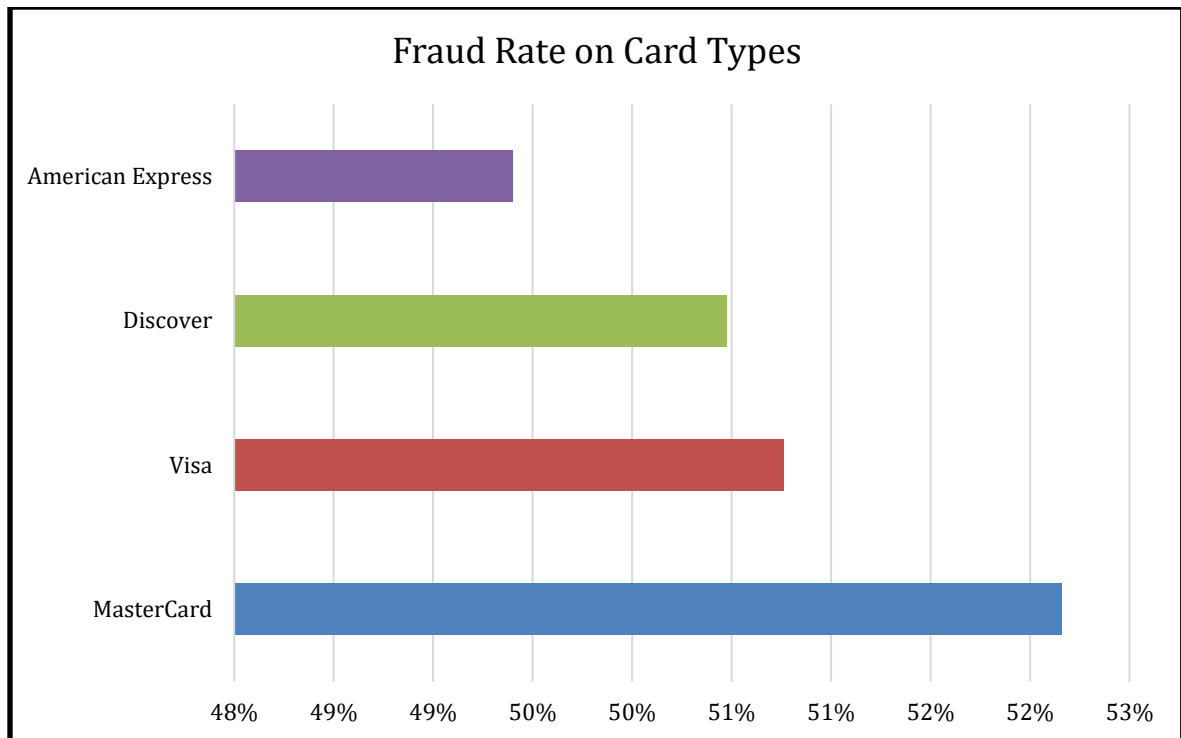
- a. How common is fraudulent transactions?
- b. Which card type are the most used?
- c. Which card type had the most amounts in transaction?
- d. Which card type has the most fraud done?

- e. Are fraudulent transactions larger?
- f. Which customer has the most fraudulent transactions?
- g. Compare fraudulent and legitimate transactions
- h. What Place of transaction has the most prominent fraud?
- i. What age does the most amount of fraud?
- j. What purchase category has the most amount of fraud?
- k. What city has the highest and the lowest fraud rate?
- l. What purchase category has the highest and lowest fraud rate?

4. Key Findings

Question a: How common is fraudulent transactions?

Fraud transactions have a 50.68% rate and non-fraud have 49.32%



Question b: Which card type are the most used?

The card type "Discover" had been used 2633 times, more than the competitors

Question c: Which card type had the most amounts in transaction?

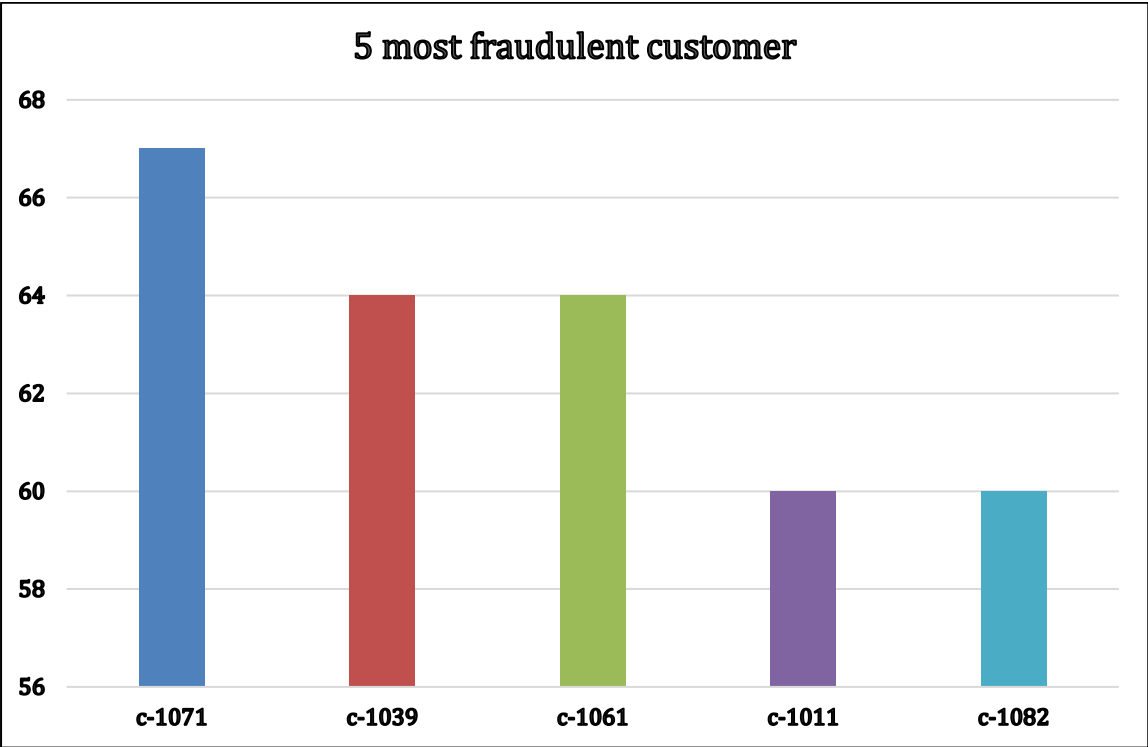
Discover made the most amount of transaction totalling in 13,319,807.23 dollars

Question d: Which card type has the most fraud done?

MasterCard has 0.521611 fraud rate, which means 52.16% transaction is fraudulent using MasterCard

Question e: Are fraudulent transactions larger?

Inferred from the data, the average amount of fraudulent transactions is larger at \$4,973.12 compared to the non-fraudulent \$4,943.22. Although both of the number only had a margin of \$30



Question f: Which customer has the most fraudulent transactions?

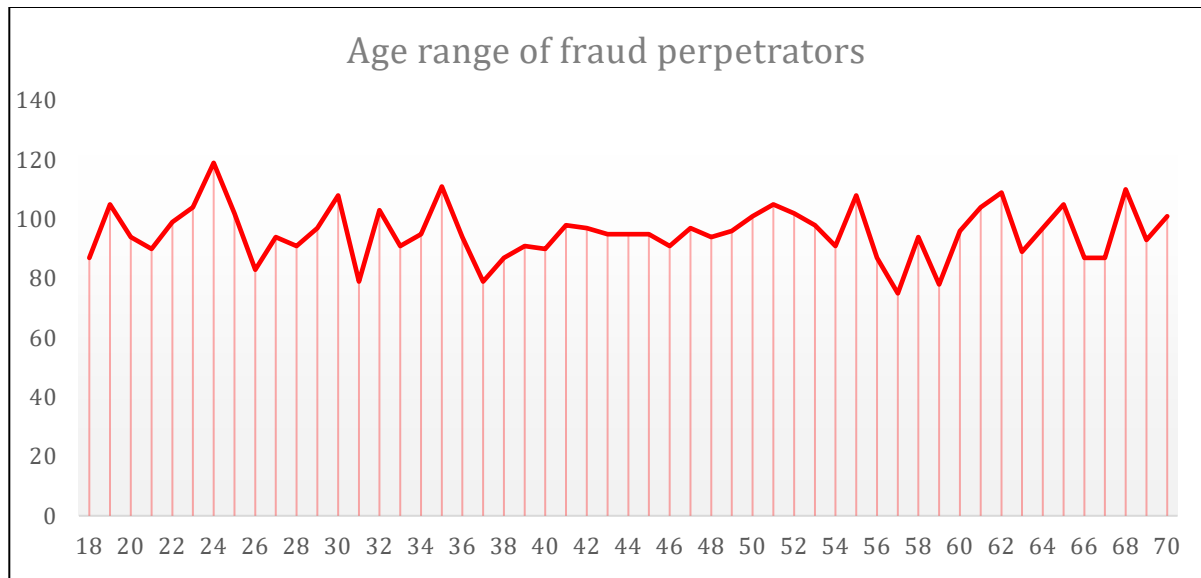
The customer ID of 1071 has a total of 67 fraudulent transactions

Question g: Compare fraudulent and legitimate transactions

The sum of fraudulent transaction is larger at 25203808.43 with a smaller median at 4922.06 and a higher frequency at 5068. For the non-fraudulent transactions, the sum is 24380007.74, with a larger median at 4958.17 and a lower frequency at 4932.

Question h: What Place of transaction has the most prominent fraud?

The place with the most prominent fraud is city-7 with 120 fraud transactions

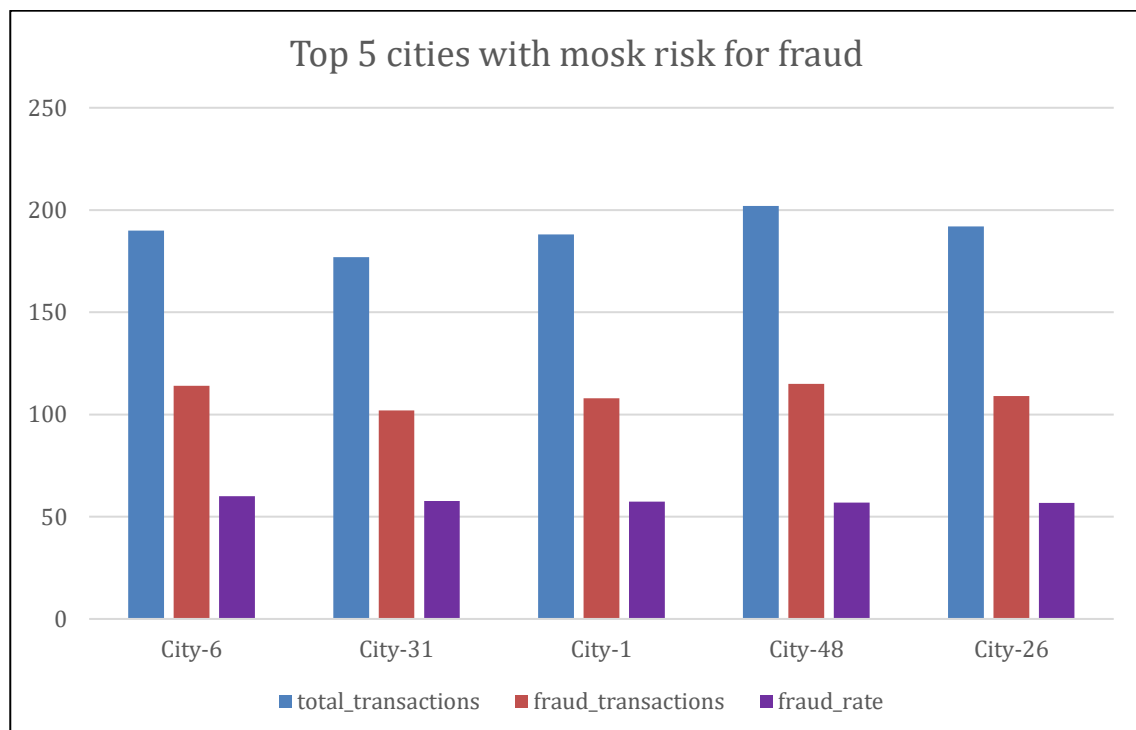


Question i: What age does the most amount of fraud?

Age that does most amount of fraud is 24 with the amount being 119 transactions

Question j: What purchase category has the most amount of fraud?

The purchase category that has the most fraud is Groceries at 896 fraud transactions



Question k: What city has the highest and the lowest fraud rate?

The city that has the highest fraud rate is City-6 with 190 total transactions and 114 fraud transactions resulting in a 60% fraud per transaction rate. The lowest

fraud rate is City-9 with 200 total transactions and 90 fraud transactions with the fraud rate of 45%

Question I: What purchase category has the highest and lowest fraud rate?

The purchase category which has the highest fraud rate is groceries with 1692 total transactions and 896 fraud transactions resulting in a 52% fraud rate. The inverse is Restaurant transactions at the lowest being 1636 total transactions with 785 fraud transactions leaving in 47% fraud transaction rate.

5. Recommendation

1. Increase monitoring of transactions originating from high-risk locations, particularly City-6, where the highest fraud rate was observed.
2. Apply enhanced fraud detection measures to grocery transactions, which exhibited both the highest fraud count and fraud rate.
3. Investigate transactions involving MasterCard to determine whether specific patterns contribute to the elevated fraud rate.
4. Flag customers with repeated fraudulent activity, such as Customer ID 1071, for further investigation.
5. Prioritize fraud rate, in addition to transaction volume, when allocating fraud monitoring resources, as high transaction counts do not necessarily indicate higher relative risk.
6. Extend future analyses with additional variables and predictive modeling to support proactive fraud detection.

*Note

The data that is used is a synthetic financial data, here is the description:

Creating a comprehensive dataset for fraud prevention and prescription in a bank involves collecting and generating various data points. In practice, data collection and generation can be a complex and time-consuming process. This dataset will be much simpler than a real-world fraud detection dataset but can serve as a starting point for a notebook. You can then expand and refine it as needed.

This Kaggle dataset contains synthetic sales data designed for data analytics practice and hackathons. The dataset is entirely computer generated and does not contain any real-world information, ensuring privacy and data protection.

Link: <https://www.kaggle.com/datasets/isabbaggin/transaction-fraudulent-financial-syntheticdata>

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